

```
!pip install --upgrade diffusers transformers -q
```

```
from pathlib import Path
import tqdm
import torch
import pandas as pd
import numpy as np
from diffusers import StableDiffusionPipeline
from transformers import pipeline, set_seed
import matplotlib.pyplot as plt
import matplotlib.pyplot as plt
import cv2
```

```
import torch
```

```
class CFG:
    device = "cuda"
    seed = 42
    image_gen_steps = 35
    image_gen_model_id = "stabilityai/stable-diffusion-2"
    image_gen_size = (900, 900)
    image_gen_guidance_scale = 9
    prompt_gen_model_id = "gpt3"
    prompt_dataset_size = 6
    prompt_max_length = 12

    @classmethod
    def initialize_generator(cls):
        if cls.device == "cuda" and torch.cuda.is_available():
            cls.generator = torch.Generator(device='cuda').manual_seed(cls.seed)
        else:
            cls.generator = torch.Generator().manual_seed(cls.seed)
```

```
# Initialize the generator
CFG.initialize_generator()
```

```
import torch
```

```
# Check if CUDA is available
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
```

```
image_gen_model = StableDiffusionPipeline.from_pretrained(
    CFG.image_gen_model_id,
    torch_dtype=torch.float,
    revision="fp16",
    use_auth_token='your_hugging_face_auth_token',
    guidance_scale=9
)
```

```
# Move the model to the device
image_gen_model = image_gen_model.to(device)
```

⚡ Cannot initialize model with low cpu memory usage because `accelerate` was not found

```
...
```

```
pip install accelerate
```

```
...
```

```
.
/usr/local/lib/python3.10/dist-packages/huggingface_hub/utils/_token.py:88: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models.
warnings.warn(
/usr/local/lib/python3.10/dist-packages/diffusers/pipelines/pipeline_loading_utils.py:
warnings.warn(
unet/diffusion_pytorch_model.safetensors not found
Keyword arguments {'use_auth_token': 'your_hugging_face_auth_token', 'guidance_scale':
Loading pipeline components...: 100%          5/5 [00:25<00:00, 5.60s/it]
```

100% 35/35 [48:42<00:00, 82.94s/it]



Double-click (or enter) to edit

✦ The Following are other examples we try to Generate Images

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100%

35/35 [48:03<00:00, 81.92s/it]



```
generate_image("astronaut in space", image_gen_model)
```



100%

35/35 [48:48<00:00, 83.74s/it]



```
generate_image("coffee with pen and laptop", image_gen_model)
```



100%

35/35 [48:46<00:00, 84.08s/it]



```
generate_image("banner for arts day", image_gen_model)
```



100%

35/35 [47:16<00:00, 81.46s/it]

